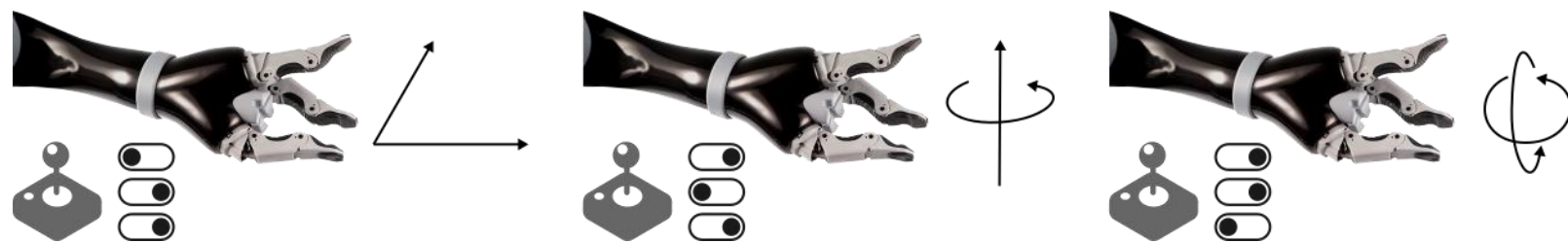


Shared Autonomy



Dylan Losey





This Lecture



- How do we share autonomy between humans and robots?
- What are some practical challenges for shared autonomy?

Shared Autonomy

Consider a system with:

- State s
- Robot action a_R
- Human action a_H

In shared autonomy the dynamics depend on human and robot actions:

$$s^{t+1} = f(s^t, a_R^t, a_H^t)$$

Shared Autonomy



Shared Autonomy



Shared Autonomy



Shared Autonomy

Key questions to consider:

- Should we just execute the human's command a_H ?
- If we are going to change it, how should we combine the robot's idea a_R with the human's action?
- What if the human and robot disagree?
- How do we determine the right assistive actions for the robot?

Shared Autonomy

Key questions to consider:

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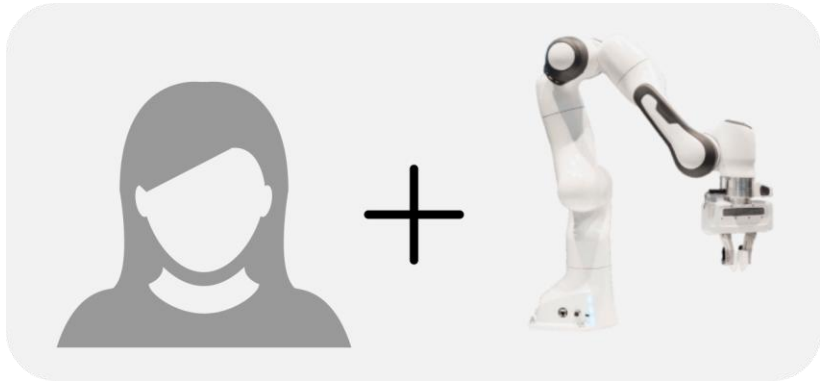
A key idea in shared autonomy is an **allocation of roles**. Intuitively, the human is in charge of the **high-level** (i.e., what task to complete), and the robot helps execute the **low-level actions** for that task



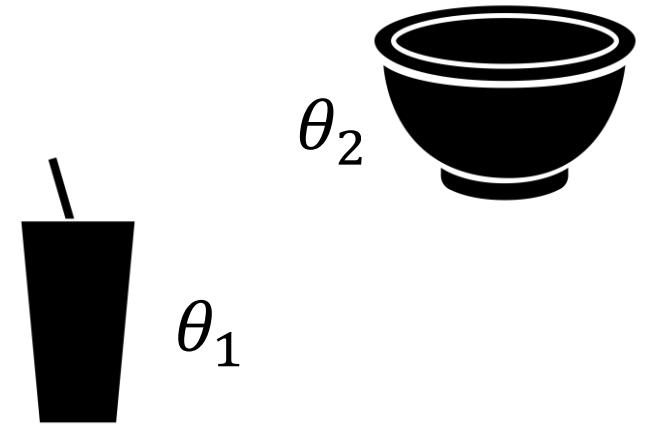
How do robots share autonomy?



Predict & Blend



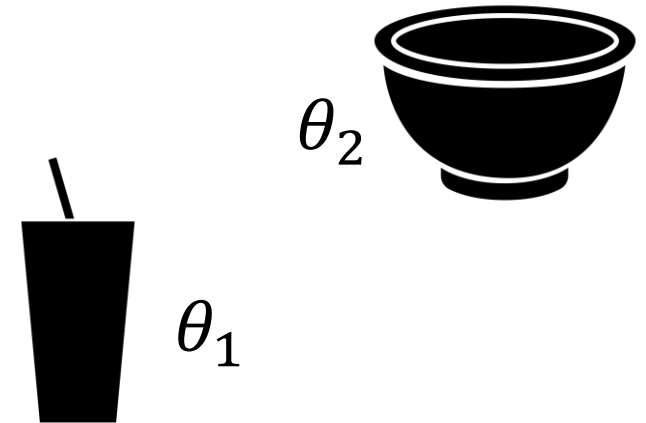
Imagine the human and robot are trying to reach for a **goal** in **free space**.



Predict & Blend

Step 1 (Predict): Infer which goal the human is trying to reach

Step 2 (Blend): Blend the human's commanded action with assistive action

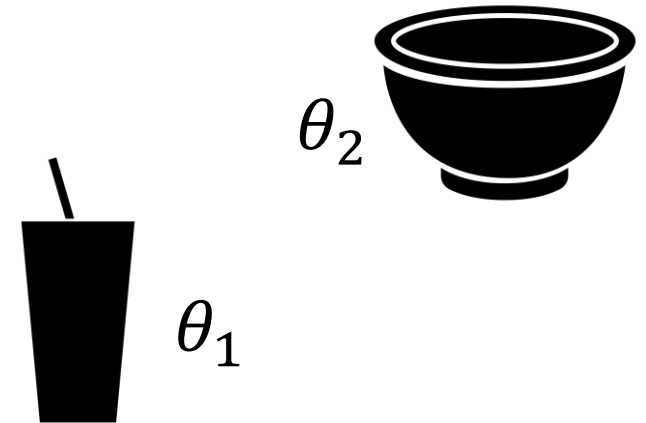


Predict & Blend

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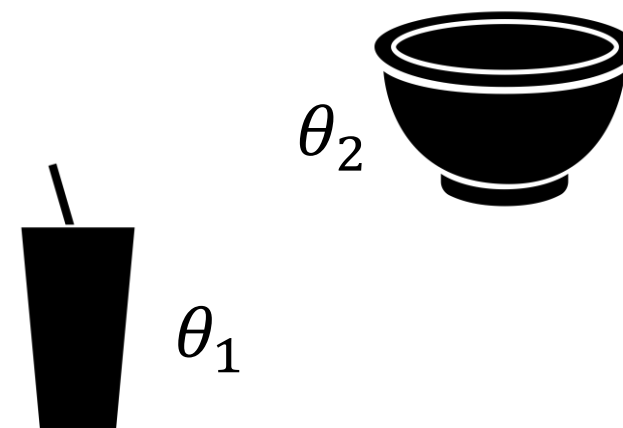
Predict & Blend

Step 1 (Predict): Infer which goal the human is trying to reach

- Start with prior $P(\theta)$
- At each timestep get (s, a_H)
- Update posterior:

$$P(\theta | D) \propto P(\theta) \prod_{(s, a_H) \in D} P(a_H | s, \theta)$$

Often people simplify this...



$$s^0 = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$



$$s^t = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\theta_2 = \begin{bmatrix} 8 \\ 3 \end{bmatrix}$$



$$\theta_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$s^0 = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$



$$s^t = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Naïve approach

Probability human wants a goal is
inverse prop. to distance from goal

$$P(\theta \mid D) \propto \frac{1}{\|\theta - s^t\|}$$



Distance left to θ



$$\theta_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$\theta_2 = \begin{bmatrix} 8 \\ 3 \end{bmatrix}$$



$$s^0 = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$



$$s^t = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Naïve approach

Probability human wants a goal is
inverse prop. to distance from goal

$$P(\theta \mid D) \propto \frac{1}{\|\theta - s^t\|}$$

$$P(\theta_2 \mid s^t) = 0.25$$



$$P(\theta_1 \mid s^t) = 0.75$$

$$s^0 = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$



$$s^t = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Better approach

Consider how efficiently human is moving towards the goal

$$P(\theta \mid D) \propto \frac{\|\theta - s^0\|}{\underbrace{\|s^t - s^0\|}_{\text{Distance gone so far}} + \underbrace{\|\theta - s^t\|}_{\text{Distance left to } \theta}}$$



$$\theta_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$\theta_2 = \begin{bmatrix} 8 \\ 3 \end{bmatrix}$$



$$s^0 = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$



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Better approach

Consider how efficiently human is moving towards the goal

$$P(\theta | D) \propto \frac{\|\theta - s^0\|}{\|s^t - s^0\| + \|\theta - s^t\|}$$

$$P(\theta_2 | s^t) = 0.52$$



$$P(\theta_1 | s^t) = 0.48$$

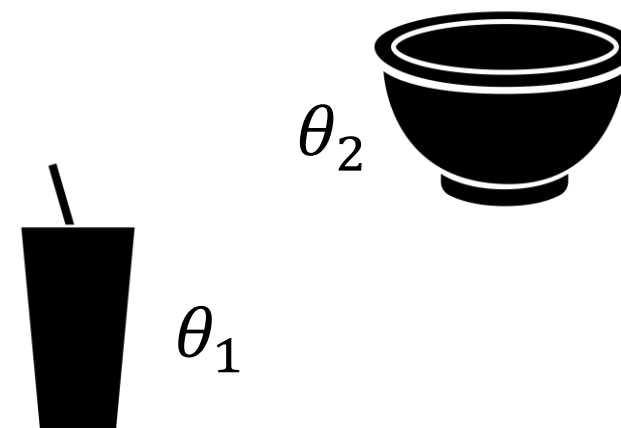
Predict & Blend

Step 1 (Predict): Infer which goal the human is trying to reach

- Start with prior $P(\theta)$
- At each timestep get (s, a_H)
- Update posterior:

$$P(\theta \mid D) \propto P(\theta) \frac{\exp(\beta \cdot \|\theta - s^0\|)}{\exp(\beta \|s^t - s^0\| + \beta \|\theta - s^t\|)}$$

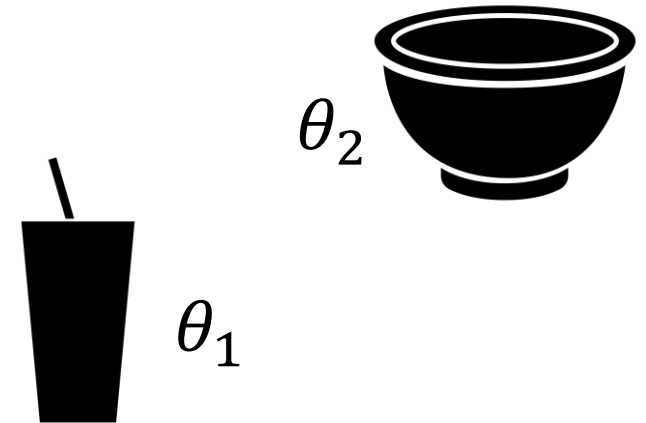
One common simplification for free space goals



Predict & Blend

Step 1 (Predict): Infer which goal the human is trying to reach

Step 2 (Blend): Blend the human's commanded action with assistive action

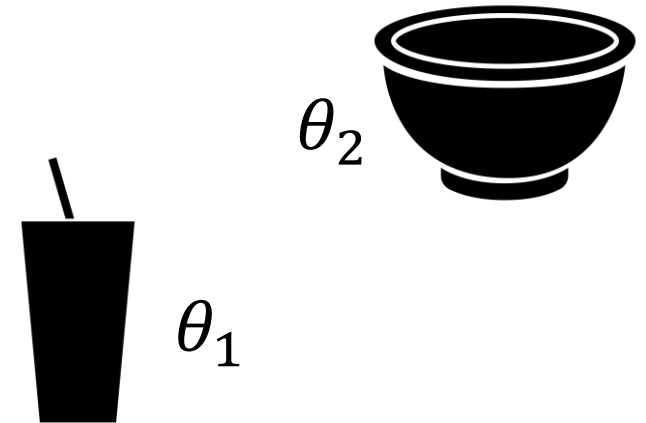


Predict & Blend

Step 2 (Blend): Blend the human's commanded action with assistive action

$$a = (1 - \alpha) \cdot a_H + \alpha \cdot a_R$$

*Linearly blend the human and robot actions,
the robot executes the overall action a*



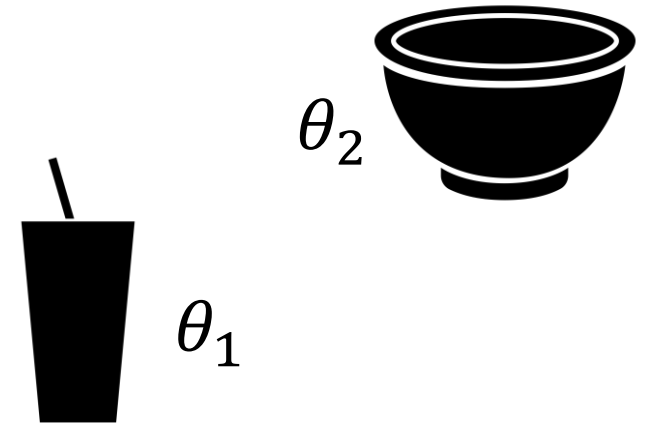
Predict & Blend

Step 2 (Blend): Blend the human's commanded action with assistive action

$$a = (1 - \alpha) \cdot a_H + \alpha \cdot a_R$$

We know this...

*What about the assistive
robot action?*

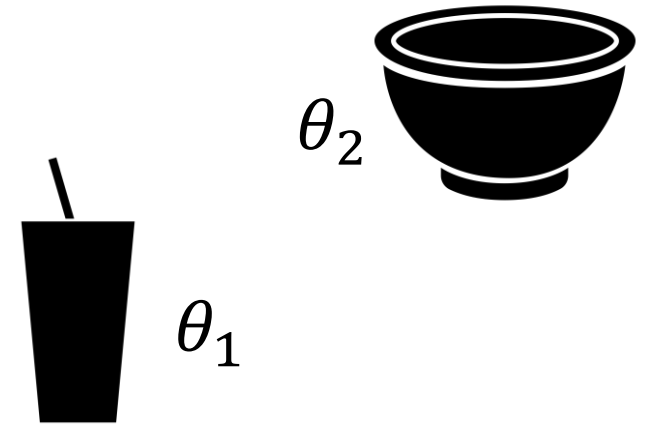


Predict & Blend

Step 2 (Blend): Blend the human's commanded action with assistive action

$$a = (1 - \alpha) \cdot a_H + \alpha \cdot a_R$$

$$a_R = \sum_{\theta \in \Theta} \underbrace{P(\theta|D)}_{\text{Assist towards weighted average goal}} \cdot (\theta - s^t)$$

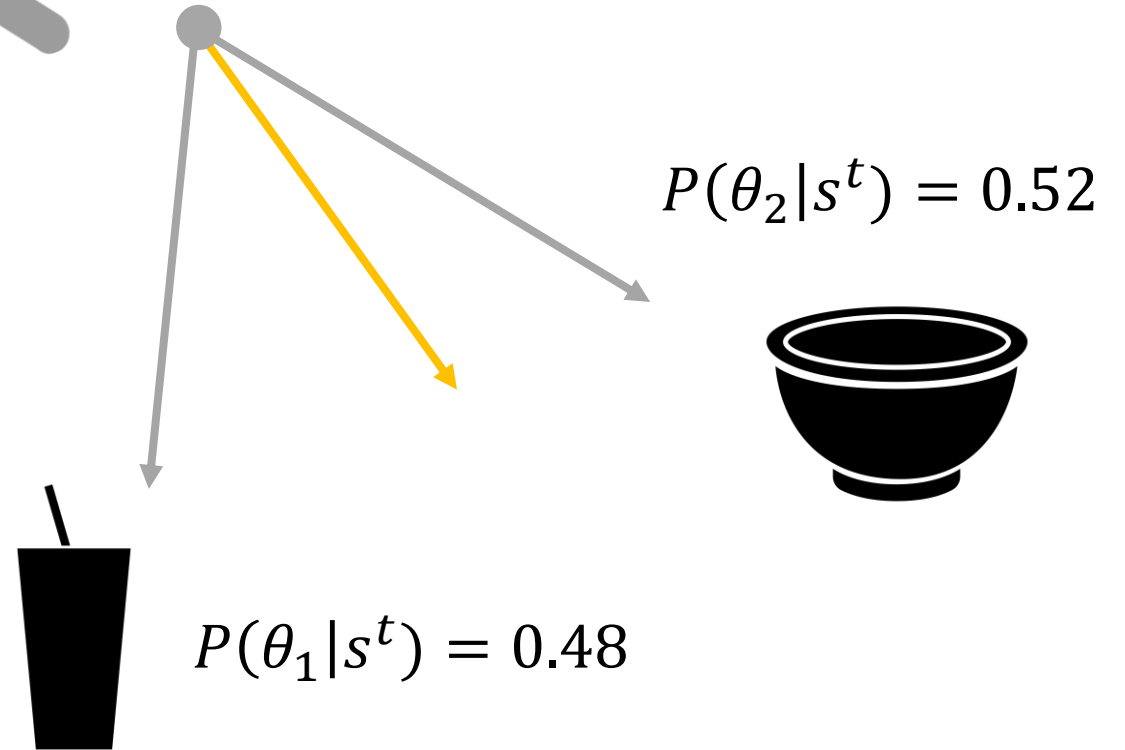


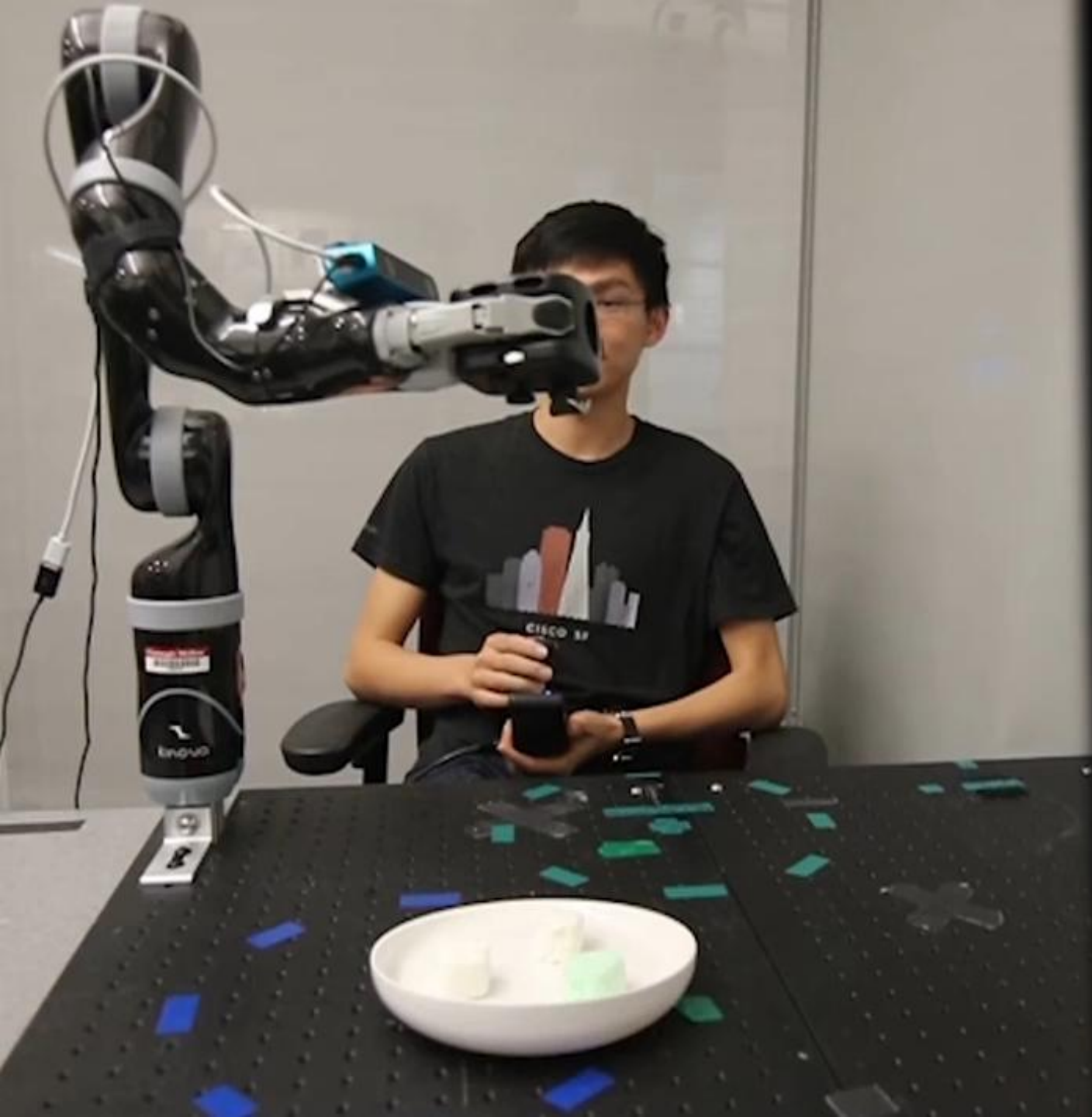


Step 2 (Blend): Blend the human's commanded action with assistive action

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Assistive arms
allow people
with disabilities
and robots to
work together to
perform tasks,
like eating.



What are some
practical challenges
for shared
autonomy?







The robot autonomy generates trajectory segments

Practical Considerations

How do humans input their actions?

Joysticks are one options, but language, eye gaze, EMG, and other modalities are also possible.
The right input often depends on the user's needs

What if the human disagrees with the robot?

We need a secondary mechanism that turns off the shared autonomy when the agents are not aligned. This could be a switch that the user toggles

What about tasks that go beyond just reaching goals?

Can use the same paradigm, but θ_1 and θ_2 become subtasks, and we try to infer which subtask the human is currently trying to perform

Practical Considerations

These practical considerations mean that there are **different types of shared autonomy algorithms** (besides Predict & Blend). Despite the differences, the unifying theme is that the human is in control, and the autonomous system is trying to help that human complete tasks

Reject Unsafe Actions

At each timestep t :

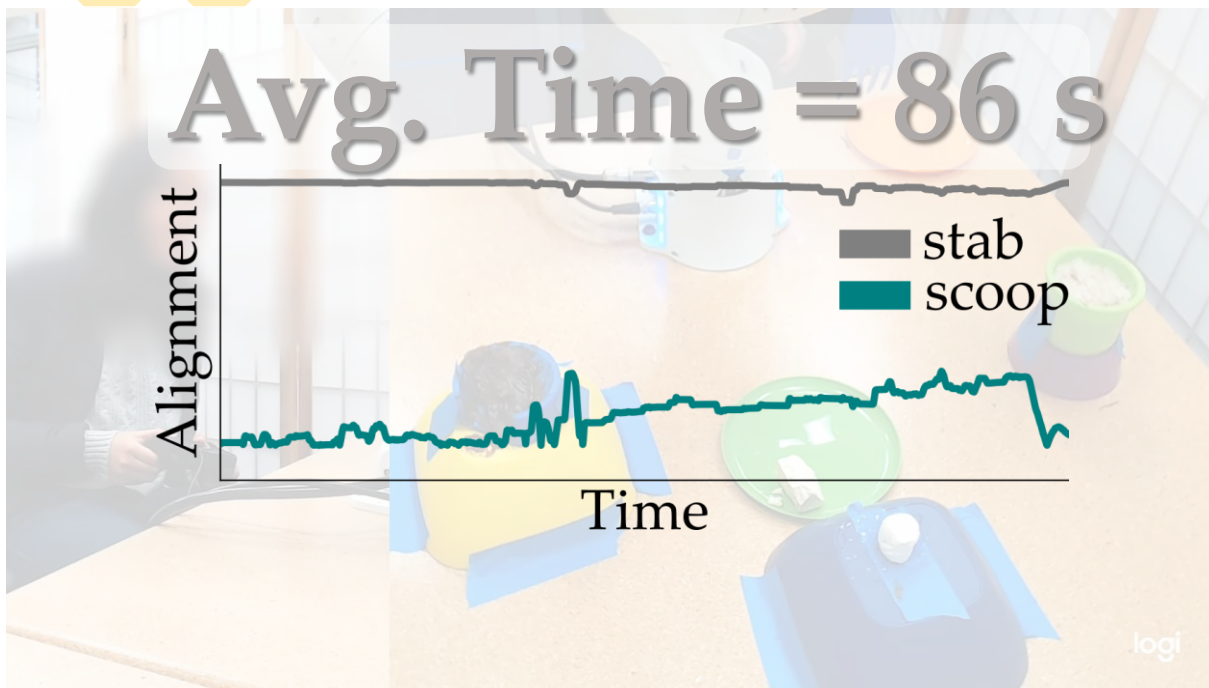
- Observe state s and human input a_H
- Measure the score $r(s, a_H)$ associated with the human's action
- If the score is above a threshold:
 - Execute human input a_H
- Otherwise:
 - Choose an action that is near to a_H but within the score threshold

Project Human Actions

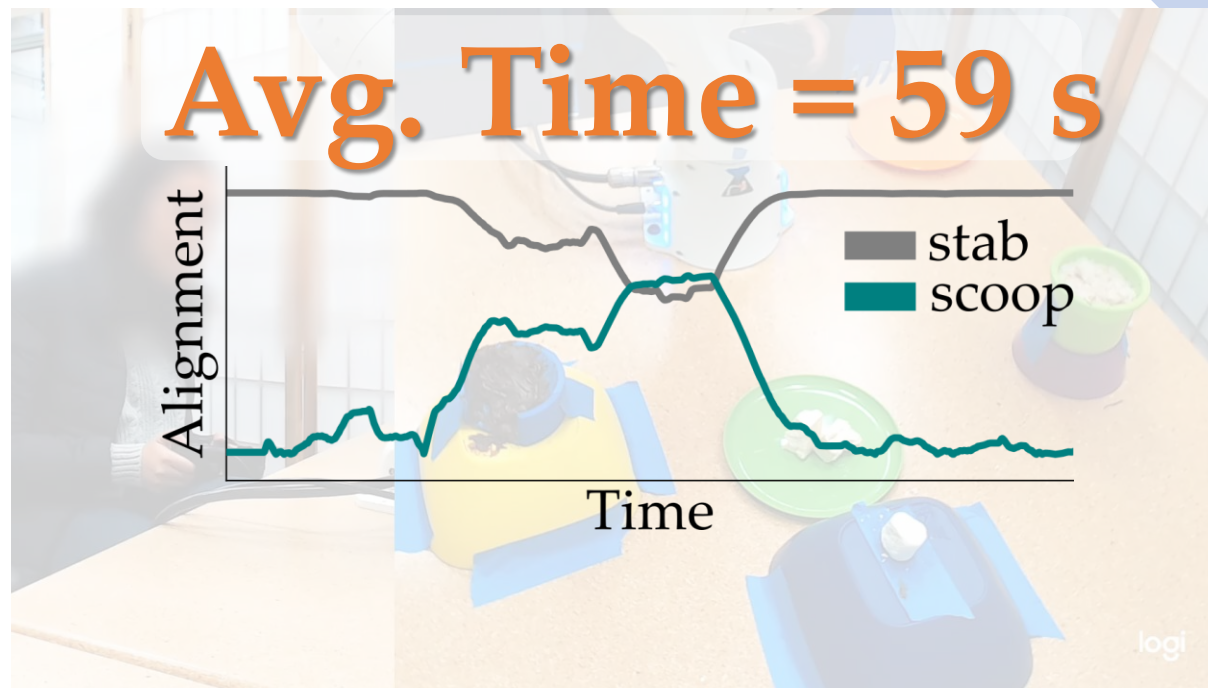
At each timestep t :

- Observe state s and human input a_H
- Estimate which subtask the human is trying to perform
- If the robot is confident about what the human wants:
 - Take the next action for that subtask
- Otherwise:
 - Execute the human's action without any assistance

You can probably think of other variations
that make sense for your scenario!



R



LA + SA (ours)

Takeaways

- In shared autonomy the human and robot share control over the robot's motion
- The robot uses a human model to infer the human's intent
- Then the robot selects low-level actions to help achieve that intent
- We can improve shared autonomy by **accurately inferring what the human wants**, by **making it easy for the human to input their intent**, and by **seamlessly resolving conflicts where the human and robot disagree**

This Lecture



- How do we share autonomy between humans and robots?
- What are some practical challenges for shared autonomy?

Next Lecture



- How do we make sure robots are safe during HRI?